Name: _____

TOPIC TEST 2

Chapter 7: Loans and investments

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 9 short-answer questions (27 marks)
- Part C: 3 analysis questions (18 marks)

Part A

Multiple-choice questions

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Sharon invests $14 000 at 5.8% p.a. simple interest, added weekly. How much will she have in 20 weeks?

A $312.31

B $14 315.64

C $14 312.31

D $16 240

E $315.64

[1 mark]

2 Sierra purchases an entertainment system worth $4500 on terms. She paid $1100 deposit and 11.8% p.a. flat rate interest on the balance. She pays the balance in equal fortnightly instalments over 1 year. The amount of each fortnightly instalment is:

A $146.20

B $193.50

C $173.08

D $130.77

E $147.11

[1 mark]

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3 The difference equation $t_{n+1} = t_n \times 1.086$ where $t_1 = 37\,000$, represents an investment of:

A \$37 000 at 1.086% p.a. compounded yearly

B \$86 000 at 37% p.a. compounded yearly

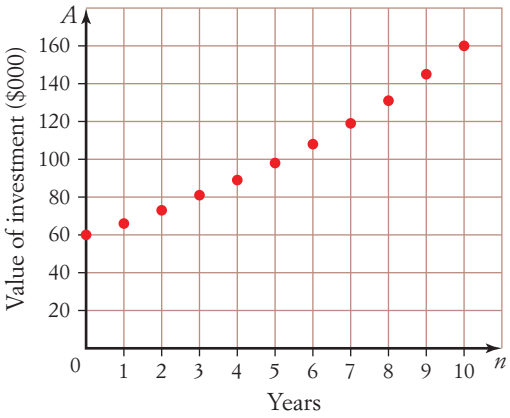
C \$37 000 at 10.86% p.a. compounded yearly

D \$37 000 at 8.6% p.a. compounded yearly

E \$1086 at 3.7% p.a. compounded yearly

[1 mark]

4 The following graph shows the increasing value of an investment over 10 years.



A rule for evaluating the value of the investment, in \$, after n years could be:

A $60 \times 0.103 \times n$

B $60\,000 \times (1.103)^n$

C $60\,000 \times 0.103 \times n$

D $60 \times (1.103)^n$

E $60\,000 \times (0.103)^n$

[1 mark]

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5 Neil and Jo each invest $27 000 at the following rates:

- Neil: 7.65% p.a. compounding quarterly
- Jo: 7.65% p.a. simple interest

At the end of 4 years, Neil will have:

A $1297.74 less than Jo

B $1297.74 more than Jo

C the same amount of money as Jo

D $997.34 less than Jo

E $997.34 more than Jo

[1 mark]

Part B

Short-answer questions

9 short-answer questions
27 marks
Show your working where appropriate.

1 Find the interest paid for a month if Arthur borrows $27 500 at 6.9% flat rate of interest.

[2 marks]

2 Renee wants to earn $5000 interest in 2 years. She invests in an account earning 11.2% p.a. simple interest. How much would she need to invest? Answer to the nearest one hundred dollars.

[3 marks]

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3 Michael purchases office furniture valued at $4800 on terms of $480 deposit and 24 monthly instalments of $224.04.

a How much interest does he pay in total?

b Calculate the annual flat rate of interest charged. Write your answer as a percentage correct to two decimal places.

[4 marks]

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4 $46 500 is invested at 6.3% p.a. compounded yearly.

a Using the formula $t_n = ar^{n-1}$, find the rule for the geometric sequence.

b What will be the value of the investment after 8 years?

[3 marks]

5 Bradley invests $19 000 in an account paying 6.6% p.a. compounded quarterly.

a Write an expression involving the compound interest formula that can be used to find the value of his investment at the end of 5 years.

b Find the value of his investment at the end of 5 years.

[4 marks]

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6 Narelle invests $16 400 in government bonds earning 7.15% p.a. compounding weekly. How much interest will she earn in 15 weeks?

[3 marks]

7 How long will $36 500 need to be invested at 7.8% p.a. compounded monthly in order to earn $7500 interest?

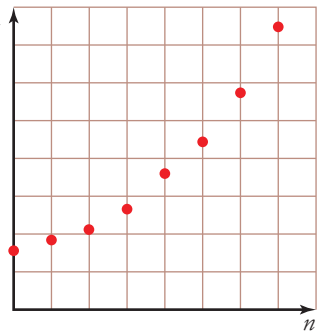
[3 marks]

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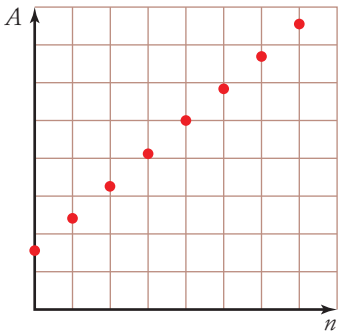
8 Patrice invests her money in an account earning 7.9% p.a. compounded monthly. She sketches a graph showing the value of her investment over 7 years.

Which of the following could represent her investment? Justify your answer.

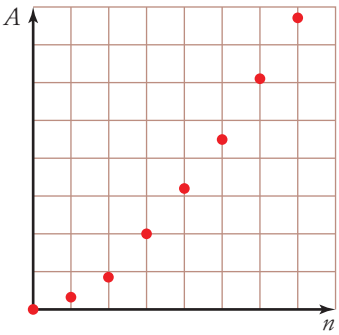
A



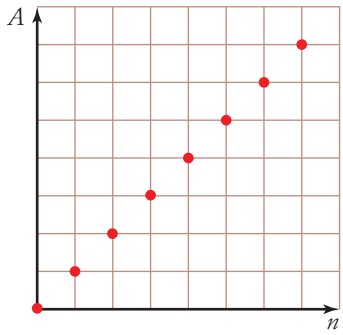
B



C



D



[3 marks]

9 Calculate the effective interest rate for an investment earning 8.65% compounded 6-monthly. Answer as a percentage correct to one decimal place.

[2 marks]

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Part C

Analysis questions

3 analysis questions
18 marks
Show your working where appropriate.

1 Melissa invests $4500 at 8.8% p.a. compounding 6-monthly. In working out the interest earned in the sixth year, she writes the following expression where she calculates the interest earned in 5 years and subtracts it from the interest earned in 6 years.

$$4500 \times 1.088^6 - 4500 \times 1.088^5$$

a Explain why this expression is incorrect.

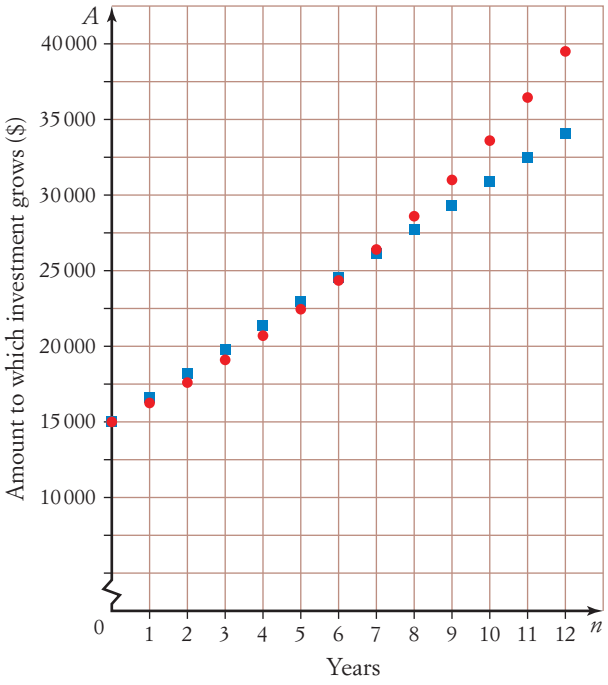
b Write the correct expression.

c Hence, calculate the interest earned in the sixth year.

[3 marks]

2 Easy Saver and Super Saver are two investment options at a bank. They both pay interest yearly, but Easy Saver pays simple interest and Super Saver pays compound interest.

The graph below shows the amount to which each investment has grown over a 12-year period when $15 000 was initially invested in both.



a How can you tell that the red-dotted line represents Super Saver?

b Which investment pays the lowest annual interest rate, Easy Saver or Super Saver? Justify your answer.

c After 12 years, the Easy Saver investment has grown to $34 100.

i Find the interest paid yearly.

ii Find the yearly interest rate for the Easy Saver investment. Answer as a percentage correct to one decimal place.

d After 12 years, the Super Saver investment has grown to $39 500. Find the yearly interest rate for the Super Saver investment. Answer as a percentage correct to one decimal place.

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e If someone was deciding between Easy Saver and Super Saver, what advice would you give them? Explain.

[10 marks]

3 Alex was looking to invest money over 5 years. He is comparing different investment accounts:

- Account 1: 9.2% p.a. simple interest.
- Account 2: 9.2% p.a. compounded annually.

a The accounts have the same interest rate per annum but Account 1 earns less interest. Explain why.

Alex was also considering another account:

- Account 3: 9.1% p.a. compounded quarterly.

b Calculate the effective interest rates for Accounts 2 and 3.

c Which is the best option for him? Explain.

[5 marks]

[Total marks: 50]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 $I = ?, P = 14\,000, r = 5.8, n = \frac{20}{52}$

$$I = \frac{Prn}{100}$$
$$= \frac{14\,000 \times 5.8 \times \frac{20}{52}}{100}$$
$$= 312.307\ldots$$
$$\approx 312.31$$
$$\therefore A = I + P$$
$$= 312.31 + 14\,000$$
$$= 14\,312.31$$

$\therefore$ **C**

2 Balance $= 4500 - 1100$
$= 3400$

$\therefore I = ?, P = 3400, r = 11.8, n = 1$

$$I = \frac{Prn}{100}$$
$$= \frac{3400 \times 11.8 \times 1}{100}$$
$$= 401.20$$

$\therefore$ Total to repay $= 401.20 + 3400$
$= 3801.20$

$\therefore$ Fortnightly instalment $= 3801.20 \div 26$
$= 146.20$

$\therefore$ **A**

3 $t_1 =$ initial value
$=$ the principal
$= 37\,000$

$$1.086 = 1 + \frac{\text{interest rate}}{100}$$
$$= 1 + \frac{8.6}{100}$$

$\therefore$ Interest rate $= 8.6\%$ p.a.

$\therefore$ **D**

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4 Graph is curved, so we are using compound interest.

Therefore, it will take the form:

$A = P\left(1 + \frac{r}{100}\right)^n$ $\therefore$ not options A or C.

Reading from the graph, the initial starting value (i.e. the principal) is \$60 000.

$\therefore$ not option D.

Value in parentheses must be >1 $\therefore$ not option E.

$\therefore$ **B**

5 Neil:

$N = 4 \times 4 = 16$

$I(\%) = 7.65$

$PV = -27\,000$

$Pmt = 0$ $\quad\quad\quad\quad$ $\therefore FV = 36\,559.735\ldots$

$FV = ?$ $\quad\quad\quad\quad\quad\quad$ $\approx 36\,559.74$

$PpY = \left.\begin{matrix}\\ \\ \end{matrix}\right\}$
$CpY = \left.\begin{matrix}\\ \\ \end{matrix}\right\} 4$

$PmtAt = END$

Jo:

$I = ?, P = 27\,000, r = 7.65, n = 4$

$$I = \frac{Prn}{100}$$
$$= \frac{27\,000 \times 7.65 \times 4}{100}$$
$$= 8262$$

$\therefore A = I + P$

$= 8262 + 27\,000$

$= 35\,262$

$\therefore$ Difference $= 36\,559.74 - 35\,262$

$= 1297.74$

$\therefore$ Neil will have \$1297.74 more than Jo.

$\therefore$ **B**

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Part B – Short-answer questions

1 $I = ?, P = 27\,500, r = 6.9, n = \frac{1}{12}$ **[1 mark]**

$$I = \frac{Prn}{100}$$

$$= \frac{27\,500 \times 6.9 \times \frac{1}{12}}{100}$$

$$= 158.125$$

$$\approx 158.13$$ **[1 mark]**

$\therefore$ \$158.13 interest paid for a week.

2 $I = 5000, P = ?, r = 11.2, n = 2$ **[1 mark]**

$$I = \frac{Prn}{100}$$

$$5000 = \frac{P \times 11.2 \times 2}{100}$$ **[1 mark]**

$$\therefore P = 22\,321.428\ldots$$

$$\approx 22\,300$$ **[1 mark]**

$\therefore$ She needs to invest \$22 300.

3 a Total paid $= 224.04 \times 24 + 480$

$= 5856.96$ **[1 mark]**

$\therefore$ Interest $= 5856.96 - 4800$

$= 1056.96$ **[1 mark]**

Therefore, he paid \$1056.96 interest.

b Amount borrowed $= 4800 - 480$

$= 4320$ **[1 mark]**

$\therefore I = 1056.96, P = 4320, r = ?, n = 2$

$$I = \frac{Prn}{100}$$

$$1056.96 = \frac{4320 \times r \times 2}{100}$$

$\therefore r = 12.233\ldots\%$

$= 5.87\%$

Therefore, the flat rate of interest charged is 12.23% p.a. **[1 mark]**

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4 a a = principal

$= 46\,500$

$r = 1 + \dfrac{\text{interest rate per time period}}{100}$

$= 1 + \dfrac{6.3}{100}$

$= 1.063$

$\therefore t_n = ar^{n-1}$

$= 46\,500 \times 1.063^{n-1}$ **[1 mark]**

b After 8 years, $n = 9$ **[1 mark]**

$\therefore t_n = 46\,500 \times 1.063^{9-1}$

$= 75\,808.703\ldots$

$\approx 75\,808.70$

$\therefore$ The value of the investment after 8 years will be \$75 808.70. **[1 mark]**

5 a $A = ?, P = 19\,000, r = \dfrac{6.6}{4} = 1.65\%$ per quarter, $n = 5 \times 4 = 20$ quarters

[1 mark for *r* and 1 mark for *n*]

$A = P\left(1 + \dfrac{r}{100}\right)^n$

$= 19\,000\left(1 + \dfrac{1.65}{100}\right)^{20}$ **[1 mark]**

Should also accept $19\,000 \times 1.0165^{20}$

b $A = 26\,357.320\ldots$

$\approx$ \$26 357.32

His investment will be worth \$26 357.32 at the end of 5 years. **[1 mark]**

6 $N = 15$ **[1 mark for writing known values either in list or in formula.]**

$I(\%) = 7.15$

$PV = -16\,400$

$Pmt = 0$ $\quad \therefore FV = 16\,741.525\ldots$

$FV = ?$ $\quad \approx 16\,741.53$ **[1 mark for *FV*]**

$PpY = \left.\begin{matrix} \\ \\ \end{matrix}\right\} 52$ $\quad \therefore I = 16\,741.53 - 16\,400$

$CpY = \left.\begin{matrix} \\ \\ \end{matrix}\right\}$

$PmtAt = END$ $\quad = 341.53$

Therefore, she will earn \$341.53 interest in 15 weeks. **[1 mark]**

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7 $N = ?$ **[1 mark for writing known values either in list or in formula.]**

$I(\%) = 7.8$

$PV = -36\,500$

$Pmt = 0$ $\therefore N = 28.8\ldots$ months **[1 mark]**

$FV = 44\,000$ ≈ 29 months

$PpY = \left.\begin{matrix}\\ \\ \end{matrix}\right\} 12$
$CpY = \left.\begin{matrix}\\ \\ \end{matrix}\right\}$

$PmtAt = END$

Therefore, it will take 2 years and 5 months. **[1 mark for *N*]**

8 Graph A **[1 mark]**

It is compound interest, so it must be a curved line graph ∴ A or C.

[1 mark for this reason]

We want the value of her investment, so, at $n = 0$, the value is the principal. Therefore, the graph must start above the zero on the y-axis ∴ A.

[1 mark for this reason]

9 $I = 8.65\% = 0.0865, \quad n = 2$

$e = (1 + \frac{i}{n})^n - 1$
$= (1 + \frac{0.0865}{2})^2 - 1$ **[1 mark]**
$= 0.0883\ldots$
$\approx 8.8\%$ **[1 mark]**

Therefore, the effective interest rate is 8.8%.

Part C – Analysis questions

1 a It has the interest rate and years as compounding yearly, but it should be compounding 6-monthly. **[1 mark]**

b $4500 \times 1.044^{12} - 4500 \times 1.044^{10}$ **[1 mark]**

c $622.516\ldots \approx 622.52$

Therefore, the interest earned in the sixth year is $622.52. **[1 mark]**

2 a It is a curved line which represents a compound interest investment. **[1 mark]**

b Super Saver **[1 mark]**

Initially, Super Saver earns the least interest. This can be seen by the fact that the line marked with dots is initially below the line marked with squares. **[1 mark]**

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c i Total interest $= 34\,100 - 15\,000$

$= 19\,100$ **[1 mark]**

$\therefore$ Interest per year $= 19\,100 \div 12$

$= 1591.666\ldots$

≈ 1591.67

$\therefore$ Interest paid per year is \$1591.67. **[1 mark]**

ii $I = 1591.67, P = 15\,000, r = ?, n = 1$ **[1 mark for using correct *I* and *P*]**

$$I = \frac{Prn}{100}$$

$$1591.67 = \frac{15\,000 \times r \times 1}{100}$$

$\therefore r = 10.61\ldots\%$ p.a.

$\approx 10.6\%$ p.a. **[1 mark]**

d $N = 12$ **[1 mark for writing known values either in list or in formula.]**

$I(\%) = ?$

$PV = -15\,000$

$Pmt = 0$ $\therefore r = 8.40\ldots$

$FV = 39\,500$ $\approx 8.4\%$ p.a.

$PpY = CpY = 1$

$PmtAt = END$

Therefore, the interest rate for the Super Saver investment is 8.4% p.a. **[1 mark]**

e If investing for more than 6 years, the Super Saver account is the best option. Otherwise, Easy Saver is the best option. This can be seen on the graph as below 7 years the Easy Saver graph is above the Super Saver but from 7 years it is reversed. **[1 mark]**

3 a Account 1 earns simple interest, which means interest is earned on the principal only. The interest earned in Account 2 is compounded, which means the interest earned each year gets added to the principal. Thus, interest is earned on the previous year's interest as well as on the principal. **[1 mark]**

b Account 2: $e = (1 + \frac{0.092}{1})^1 - 1 = 0.092\ldots$

$\approx 9.2\%$ **[1 mark]**

Account 3: $e = (1 + \frac{0.091}{4})^4 - 1 = 0.09415\ldots$

$\approx 9.41\%$ **[1 mark]**

c Account 3 is the best option. **[1 mark]**

It has the higher effective interest rate and will therefore pay higher interest over time. **[1 mark]**

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